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Web Systems Lab Portfolio Report

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Course: Web Systems

Professor/Instructor: Mr. Guillermo Red

1. Why I Built This Project

Throughout my Web Systems lab classes, I completed 11 different laboratory assignments. My work started with simple static HTML pages, moved on to browser storage and DOM manipulation, and eventually led to building web pages that pull real-time data from external APIs.

As the weeks went on, managing 11 separate .html files across different folders became messy. If I wanted to show my work to my instructor or classmate, I had to open individual files manually or send random links.

To clean things up, I decided to build a single landing page—my **Portfolio Hub**. I designed it to serve as a central dashboard where all my lab submissions are displayed as clean, interactive cards. Anyone viewing my portfolio can click on any exercise, and it opens up instantly in a new tab.

2. My Tech Stack & Code Structure

I chose to keep my setup lightweight, fast, and easy to maintain:

- **HTML5:** I used semantic tags like <header>, <main>, and <footer> to keep my page structure clean and readable.
- **Bootstrap 5.3:** I integrated Bootstrap via CDN to handle the responsive grid system, making sure my portfolio looks good whether opened on a phone, tablet, or laptop.
- **Custom CSS:** I wrote my own CSS rules to handle hover animations and custom badge colors so the interface wouldn't look like a standard Bootstrap template.

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| HEADER: BORJE, BRIAN VIEL - WEB SYSTEMS LAB SUBMISSIONS |

| Subtitle: Select any laboratory exercise below to open in a new tab. |

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| |

| [Category 1 Badge] [Category 2 Badge] [Category 3 Badge] |

| Index Week-1.html Index Week-2.html Index Week-3.html |

| [Open Lab ->] [Open Lab ->] [Open Lab ->] |

| |

| [Category 4 Badge] [Category 5 Badge] [Category 6 Badge] |

| Index Week-4.html Index Week-5.html LOCAL_STORAGE.html |

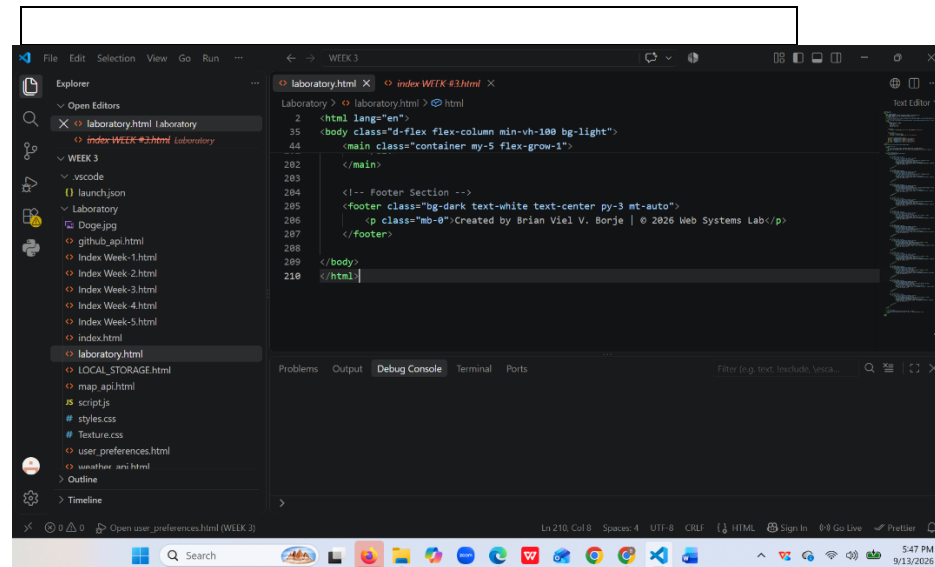
| [Open Lab ->] [Open Lab ->] [Open Lab ->] |

| |

| (Grid automatically rearranges cards depending on screen size) |

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| FOOTER: Created by Brian Viel V. Borje | © 2026 Web Systems Lab |



3. Key Design Choices & Features I Added

Interactive Hover Effects

I didn't want my portfolio to feel like a static list of links. To give visitors instant visual feedback, I added a quick CSS transition that lifts the card slightly when hovered over:

CSS

```
.lab-card {  
    transition: transform 0.2s ease-in-out, box-shadow 0.2s ease-in-out;  
}  
  
.lab-card:hover {  
    transform: translateY(-5px);  
    box-shadow: 0 10px 20px rgba(0, 0, 0, 0.15) !important;  
}
```

Custom Badge Colors

Bootstrap includes built-in badge colors like primary (blue), success (green), and danger (red). Because I had 11 assignments, repeating the same four default colors looked repetitive. I wrote custom CSS classes (.bg-purple, .bg-teal, .bg-indigo, and .bg-orange) so each category stands out with its own color.

Fixing the Floating Footer

While building the layout, I noticed the footer would float in the middle of the screen on devices with tall display resolutions. To fix this, I applied flexbox layout utilities directly to the <body>:

- min-vh-100: Ensures the body spans at least 100% of the screen height.
- flex-grow-1: Tells the <main> content container to expand and take up remaining space.
- mt-auto: Automatically pushes my footer down to the very bottom of the screen.
-

4. Summary of My 11 Laboratory Exercises

Here is what I built across all 11 exercises linked inside my hub:

Phase 1: Core Web Structure (Labs 1–5)

- **Lab 1 (Index Week-1.html):** Built my foundational HTML document using semantic tags, headers, and text formatting.
- **Lab 2 (Index Week-2.html):** Applied core CSS rules, focusing on typography, color palettes, and margin/padding mechanics.

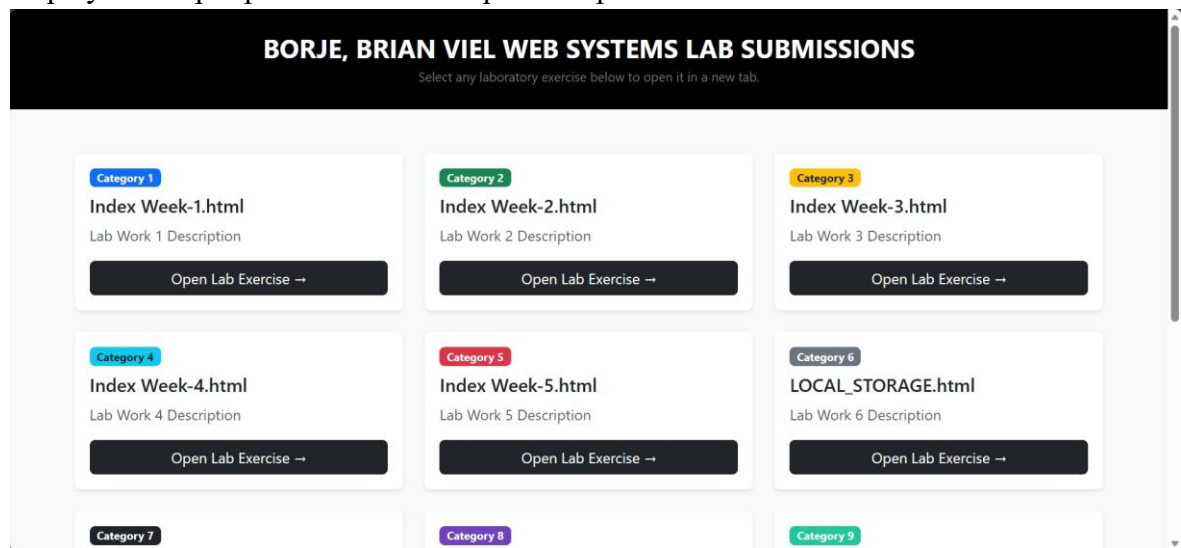
- **Lab 3 (Index Week-3.html):** Created interactive forms using input types, select dropdowns, and client-side validation rules.
- **Lab 4 (Index Week-4.html):** Handled responsive media elements and fluid image layouts for mobile screens.
- **Lab 5 (Index Week-5.html):** Built multi-column dynamic layouts using CSS Flexbox and CSS Grid.

Phase 2: Client Logic & Data Persistence (Labs 6–8)

- **Lab 6 (LOCAL_STORAGE.html):** Saved and parsed structured JSON data locally in the browser using JavaScript's localStorage.
- **Lab 7 (user_preferences.html):** Built a theme switcher (Light/Dark mode) that saves user settings across sessions.
- **Lab 8 (Index Week-8.html):** Added, removed, and edited DOM elements dynamically using JavaScript event listeners.

Phase 3: Working with Real APIs (Labs 9–11)

- **Lab 9 (weather_api.html):** Connected to the OpenWeatherMap API using fetch() and async/await to show real-time weather updates.
- **Lab 10 (map_api.html):** Integrated an interactive map interface with custom markers and geographical coordinate tracking.
- **Lab 11 (github_api.html):** Built an app that queries the GitHub REST API to fetch and display developer profile details and public repositories.



5. Security Detail I Implemented

Because every card button opens a project in a new tab (`target="_blank"`), I made sure to protect my site against **Tab-Nabbing** (a vulnerability where a newly opened tab can gain control over the original page via JavaScript's `window.opener`).

To prevent this, I added `rel="noopener noreferrer"` to all my links:

HTML

```
<a href="weather_api.html" target="_blank" rel="noopener noreferrer" class="btn btn-dark mt-auto w-100">
```

Open Lab Exercise →

```
</a>
```

- **noopener** prevents the new page from accessing my main page context.
- **noreferrer** keeps my browser requests private by hiding referrer header data.

6. What I Learned

Building this portfolio hub was a rewarding experience. It helped me organize my term work while letting me practice foundational front-end skills—like mobile-first grid layouts, CSS micro-interactions, clean HTML semantics, and simple security standards. Turning 11 separate lab exercises into a single unified web application brought everything I learned together into one project I can easily share.